QP Code:

Reg. No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) THEORYEXAMINATION OCTOBER 2026
SEMESTER V
DSC–MG5DSCPHY302–ATOMIC AND MOLECULAR PHYSICS
(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

1.	What is an absorption spectrum? Explain how it is produced.	[U] / [CO1]
2.	State Pauli's exclusion principle	[U] / [CO2]
3.	Write a note on spectral terms	[U] / [CO3]
4.	Write down the difference between normal and Anomalous Zeeman effect	[U] / [CO4]
5.	A radio wave has a wavelength of 300 m. Calculate its frequency.	[A] / [CO4]
6.	Explain visible and ultraviolet regions in the electromagnetic spectrum.	[U] / [CO4]
7.	What are Stokes and Anti-Stokes lines?	[U] / [CO5]
8.	Explain the factors affecting the intensity of Raman lines.	[An] / [CO5]
9.	State Beer Lambert law	[U] / [CO4]

10.	What is chemical shift? Explain it with a suitable example	[U] / [CO5]
11.	List any three applications of Mossbauer spectroscopy	[U] / [CO5]
12.	Explain the basic components of ESR	[U] / [CO5]

[3x10 = 30]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **8** marks

13.	With a neat diagram explain Stern Gerlach experiment	[U] / [CO1]
14.	Briefly explain vector atom model	[U] / [CO2]
15.	Explain the fine structure of sodium D lines	[U] / [CO3]
16.	How molecules are classified on the basis of their principal moments of inertia?	[U] / [CO3]
17.	Draw a neat labelled diagram of electromagnetic spectrum	[U] / [CO4]
18.	Briefly explain the classical and quantum theory of Raman effect	[U] / [CO5]
19.	Explain the principle of NMR spectroscopy	[U] / [CO5]
20.	Briefly explain ESR spectroscopy	[U] / [CO5]

[8x5 = 40]